

Deepak Sathyan

arXiv | InspireHEP | Google Scholar | ORCID | Website

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📍 Mitchell Institute for Fundamental Physics and Astronomy, Texas A&M University, College Station, TX

Education

University of Maryland

Ph.D. in Physics

Thesis: Unifying Searches for New Physics with Precision Measurements of the W Boson Mass

Advisor: Dr. Kaustubh Agashe

College Park, MD

Aug 2018 – July 2024

Boston University

Bachelor of Arts in Physics, Summa Cum Laude with Honors

GPA: 3.9

Minor: Mathematics

Thesis: Muon ($g-2$): Searching for the Muon Electric Dipole Moment

Advisor: Dr. Robert Carey

Boston, MA

Aug 2014 – May 2018

Experience

Postdoctoral Research Associate

Mitchell Institute for Fundamental Physics and Astronomy
Texas A&M University

College Station, TX

Sept 2024 – Aug 2027

3 years

Publications

Producing the GeV Galactic Center Excess via Cosmic Ray-Dark Matter Scattering

B. Dutta, D. Goswami, J. Kumar, M. Rai, D. Sathyan

arxiv.org/abs/2605.08010

May 2026

A Baryon and Lepton Number Violation Model Testable at the LHC

A. Bhoonah, F. Burk, D. Liu, T. Ou, D. Sathyan

arxiv.org/abs/2508.21064

Aug 2025

New Constraints on Neutrino-Dark Matter Interactions: A Comprehensive Analysis

P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang

arxiv.org/abs/2507.01000

July 2025

New laboratory constraints on neutrinophilic mediators

P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang

arxiv.org/abs/2407.12738 (Phys. Lett. B)

2025

‘Unification’ of BSM searches and SM measurements: the case of lepton+MET and m_W	2025
K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, D. Sathyan arxiv.org/abs/2404.17574 (JHEP)	
A new purpose for the W-boson mass measurement: Searching for New Physics in lepton+MET	2024
K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, D. Sathyan arxiv.org/abs/2310.13687 (Phys. Lett. B)	
Unifying Searches for New Physics with Precision Measurements of the W Boson Mass	2024
D. Sathyan drum.lib.umd.edu/items/f9b081c8-444a-4af2-b69e-c8156d11a2c9 (Ph.D. Thesis)	
Energy-peak based method to measure top quark mass via B-hadron decay lengths	2023
K. Agashe, S. Airen, R. Franceschini, J. Incandela, D. Kim, D. Sathyan arxiv.org/abs/2212.03929 (JHEP)	
TF07 Snowmass Report: Theory of Collider Phenomena	Oct 2022
F. Maltoni et al. arxiv.org/abs/2210.02591	
Report of the Topical Group on Physics Beyond the Standard Model at Energy Frontier for Snowmass 2021	Sept 2022
T. Bose et al. arxiv.org/abs/2209.13128	
Report of the Topical Group on Top quark physics and heavy flavor production for Snowmass 2021	Sept 2022
K. Agashe et al. arxiv.org/abs/2209.11267	
Snowmass2021 - White Paper, Implications of Energy Peak for Collider Phenomenology: Top Quark Mass Determination and Beyond	Apr 2022
K. Agashe, S. Airen, R. Franceschini, D. Kim, D. Sathyan arxiv.org/abs/2204.02928	
Snowmass2021 White Paper: Collider Physics Opportunities of Extended Warped Extra-Dimensional Models	Mar 2022
K. Agashe, J.H. Collins, P. Du, M. Ekhaterachian, S. Hong, D. Kim, R.K. Mishra, D. Sathyan arxiv.org/abs/2203.13305	
The straw tracking detector for the Fermilab Muon g-2 Experiment	2022
B.T. King et al. arxiv.org/abs/2111.02076 (JINST)	

- Measurement of the Positive Muon Anomalous Magnetic Moment to 0.46 ppm** 2021
 Muon g-2 Collaboration
arxiv.org/abs/2104.03281 (Phys. Rev. Lett.)
- Beam dynamics corrections to the Run-1 measurement of the muon anomalous magnetic moment at Fermilab** 2021
 Muon g-2 Collaboration
arxiv.org/abs/2104.03240 (Phys. Rev. Accel. Beams)
- LHC Signals for KK Graviton from an Extended Warped Extra Dimension** 2020
 K. Agashe, M. Ekhaterachian, D. Kim, D. Sathyan
arxiv.org/abs/2008.06480 (JHEP)

Articles in Preparation

- LHC as a Beam Dump: Probing Light Dark Photons** Sept 2026
 B. Dutta, A. Karthikeyan, D. Kim, H. Kim, T. Kim, D. Sathyan
arxiv.org/a/sathyan_d_1
- New Probes of Dark Sectors at a Muon Collider** Jan 2027
 B. Dutta, A. Karthikeyan, D. Kim, K. Kelly, D. Sathyan
arxiv.org/a/sathyan_d_1
- Exploring open systems approach to neutrino oscillations** Mar 2027
 A. Chandra Shekar, K. Kelly, D. Sathyan, L. Strigari, T. Zhou
arxiv.org/a/sathyan_d_1
- LHC Signals for KK Gravitons Producing 8-Particle Final States** Mar 2027
 K. Agashe, D. Kim, P. Maksimovic, S. Mondal, M. Osherson, K. Panchal, D. Plotnikov, D. Sathyan
arxiv.org/a/sathyan_d_1

Seminars

- Unification of Searches and Measurements: Probing BSM with the W boson mass measurement**
 CMS Exotica general meeting, May 2024
- A New Purpose A New Purpose for the W-mass Measurement: Searching for New Physics via $l + MET$**
 Theory Seminar at Washington University in St. Louis, October 2023
 HEP Seminar at Northwestern University, October 2023
 HEP Seminar at University of Pittsburgh, November 2023
 CMS SUSY general meeting, November 2023
 Seminar at UT Austin, December 2023
 Seminar at Texas A&M, December 2023

Conference Talks

Can the LHC be sensitive to the Light Dark World?

Light Dark World 2026 at Carleton University, July 2026

Producing the GeV Galactic Center Excess via Cosmic Ray-Dark Matter Scattering

CETUP* Workshop, June 2026

A Baryon and Lepton Number Violation Model Testable at the LHC

Particle Physics on the Plains at the University of Kansas, November 2025

Can the LHC be sensitive to light dark mediators?

Phenomenology Conference at the University of Pittsburgh, May 2025

CETUP* Workshop, June 2025

A comprehensive analysis of supernova neutrino-dark matter interactions

Mitchell Conference at Texas A&M University, May 2024

NuFact Conference, September 2024

Texas TACOS @ UT Austin, October 2024

Particle Physics on the Plains at the University of Kansas, November 2024

A New Purpose for the W-mass Measurement: Searching for New Physics via $I + MET$

Particle Physics on the Plains at the University of Kansas, October 2023

Probing Dark Matter-Neutrino Interactions via Supernova Neutrinos

Phenomenology Conference at the University of Pittsburgh, May 2023

Model-Independent Measurement of Top Quark Mass Using B-Hadron Decay Lengths (Part I)

Phenomenology Conference at the University of Pittsburgh, May 2022

Signals of KK graviton from extended warped extra dimensions at the LHC (II)

Phenomenology Conference at the University of Pittsburgh, May 2020

April APS Meeting, April 2021

Conferences Organized

The Mitchell Conference on Collider, Dark Matter, and Neutrino Physics 2026

Texas A&M University, May 27-30, 2026

The Mitchell Conference on Collider, Dark Matter, and Neutrino Physics 2025

Texas A&M University, May 13-16, 2025

Awards

Breakthrough Prize in Fundamental Physics

Muon g-2 Collaboration

Apr 2026

Schools Attended

GGI Lectures on the Theory of Fundamental Interactions

Jan 2023 – Jan 2023

TASI 2024: The Frontiers of Particle Theory

June 2024 – June 2024

Outreach

BU Student Mentorship Program PRISM (2016-2018)

Mentored incoming freshman undergraduates in physics, providing guidance on courses, research opportunities, and resources

TAMU Physics and Engineering Festival (2025-2027)

Presented demonstrations of fluid dynamics to the general public, including high school students and families

Skills

Programming: Mathematica, Python, C++

Physics Software: MadGraph5_aMC@NLO, Pythia6, Pythia8, Delphes, ROOT, GEANT4, FeynRules